



Approach to the 1-propanol dehydration using an extractive distillation process with ethylene glycol



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ABSTRACT

The extractive distillation process exploits the capacity of some chemicals to alter the relative volatility between the components of a mixture. In this way, a third component (called entrainer) may be added to an azeotropic binary mixture to break the azeotrope.

This paper discusses the potential use of ethylene glycol as entrainer in a 1-propanol dehydration process by extractive distillation. First, the present work focuses on the acquisition of isobaric vapor–liquid equilibrium data of the ternary system 1-propanol + water + ethylene glycol system and the binaries systems 1-propanol + ethylene glycol and water + ethylene glycol. All measurements were done at 101.3 kPa.

The experimental data were correlated by the Wilson, NRTL and UNIQUAC local composition models in order to obtain a set of thermodynamic parameters. Finally, the extractive distillation process was simulated and later economically optimized with the aid of commercial software (Aspen Hysys®) using the previously obtained parameters of the NRTL model.

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1. Introduction

A growing demand for energy and the negative impact that fossil fuels have on the environment make it imperative, the search for other sources of energy. One of the proposals is the use of alcohols as biofuels derived from renewable sources [1]. Among these alcohols is the 1-propanol, which can be converted into diesel fuel by esterification reaction [2,3]. However, 1-propanol dehydration is not easy because 1-propanol and water form an azeotrope that prevents the use of a conventional distillation process. In these cases, other techniques such as the extractive distillation (ED) are required to accomplish the separation of the mixture components. The ED consists of the addition of a third component to the original mixture in order to break the azeotrope and make feasible the separation [4]. This third component is known as entrainer. A typical ED process design consists in a first distillation column where the desired product is obtained as the top stream [5]. The bottom stream is usually formed by the entrainer and the second component of the original mixture. This stream is then fed to a second distillation column where the entrainer is separated and recycled.

All the designs of the extractive distillation process require thermodynamic models to correctly simulate the different distillation columns of the process. Each of these models uses different parameters to estimate the vapor–liquid equilibrium (VLE) between liquid and vapor phases present in the distillation column. There are other methods such as UNIFAC [6] to estimate these parameters theoretically. However, these predictive methods make use of sub-parameters that are calculated from experimental data. Therefore, measurement of equilibrium data becomes inevitable.

Once the model parameters are defined, simulation of the process can be carried out by means of a computer simulator. These programs allow easy calculation of all the variables involved in the process.

The present work is focused on the separation of a mixture of water and 1-propanol by extractive distillation. This paper continues our research in separation of alcohol and water mixtures using solvents with a low environment impact. Previously, 2-methoxyethanol and *n*-propyl acetate were proposed as solvents for the 1-propanol dehydration process [7,8]. In this case the selected entrainer is ethylene glycol. This chemical has been proposed by several authors [9–12] as entrainer in the extractive distillation of ethanol and water. The main features in favor of ethylene glycol are its low toxicity and reduced vapor pressure.

The first stage was the gathering of isobaric VLE data of the ternary system 1-propanol (1) + water (2) + ethylene glycol (3) and

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its constituent binary systems at 101.3 kPa: 1-propanol (1) + ethylene glycol (3) and water (2) + ethylene glycol (3). Experimental VLE data of the ternary system have not been published previously. On the other hand, binary VLE data of 1-propanol (1) + ethylene glycol (3) and water (2) + ethylene glycol (3) systems are available in the literature [13–15]. Reported data have been compared with those obtained in this work. VLE data from 1-propanol (1) + water (2) were not experimentally determined because binary interaction parameters were obtained elsewhere [7]. All binary and ternary experimental data were found thermodynamically consistent after successfully passing the test of Fredenslund et al. [16] and the Wisniak and Tamir [17] modification of the McDermott–Ellis test [18], respectively.

The next stage was correlating binary VLE data by minimizing an objective function to obtain the parameters of the Wilson, NRTL and UNIQUAC local composition models. After estimating the ternary VLE data using each of these models, the best model was selected to use in the simulation of extractive distillation process. In this way, an extractive distillation process using ethylene glycol as entrainer was simulated and optimized with the aid of the commercial software Aspen Hysys[®].

2. Experiments

2.1. Chemicals

1-Propanol (>0.990 mass fraction) was supplied by Sigma–Aldrich and both water (>0.995 mass fraction, water for chromatography LiChrosolv) and ethylene glycol (>0.995 mass fraction) were supplied by Merck. No significant impurities were found after analyzing all chemicals by chromatography. As a result, no purification methods were applied. A Karl Fischer volumetric automatic titrator (Metrohm, 701 KF Titrino) was used to determine the water content. The results showed a negligible amount of water in pure organic components. Anyway, both chemicals were dried over molecular sieves (Aldrich, type 4 Å, 1.6 mm pellets). Table 1 shows a summary of the chemicals employed in this work.

2.2. Apparatus and procedure

The equilibrium vessel used to carry out the experiments was the Labodest VLE 602/D which is an all-glass dynamic-recirculating still. The main feature of this apparatus is its Cottrell circulation pump, which ensures a proper contact between liquid and vapor phases. The device is manufactured by Fischer Labor und Verfahrenstechnik (Germany) and several references to it can be found in literature [19–21]. A digital Hart Scientific thermometer model 1502 A and a Pt 100 probe Hart Scientific model 5622 were used to measure the equilibrium temperature. The probe was previously calibrated at the ENAC-accredited Spanish Instituto Nacional de Técnica Aeroespacial and it was checked using the ice and steam points of distilled water. The standard uncertainty is estimated to be 0.01 K. A Fischer M 101 pressure control system allows keeping equilibrium pressure at 101.3 kPa with an estimated value of the standard uncertainty equal to 0.1 kPa. The

manometer was calibrated with vapor pressure of water (>0.995 mass fraction).

In all experiments the pressure was set at 101.3 kPa. This value is held constant throughout the experiment by using the pressure controller together with a vacuum pump which is connected to the equilibrium apparatus through an electronic valve. Then, the heating and stirring systems of the liquid mixture were started. After several minutes (about 25 or 30 min) equilibrium was reached. The main evidence that equilibrium has been reached is the stability of the values of temperature and pressure. The last step was to analyze several samples of condensed vapor and liquid. Special syringes to withdraw small volume samples (0.5 μL) were used to carry out the sampling. The experiment ended when two consecutive samples of the liquid phase had the same composition (with a difference of ±0.001 mole fraction) and two consecutive samples of the vapor phase had also the same composition (with a difference of ±0.001 mole fraction).

2.3. Analysis

Analyses were conducted by gas chromatography (GC). The chromatograph used in this work was a CE Instruments GC 8000 Top which employs a thermal conductivity detector (TCD) and an 80/100 Porapak Q 3 m × 1/8 in. column. The response peaks were processed with the Chrom-Card for Windows software. The operating conditions in GC were an injection temperature of 373 K, a detector temperature of 433 K and an oven temperature of 373 K during the first five minutes and 433 K in the rest of the analyzing time. The detector current was 220 mA and the helium flow rate was established in 40 mL/min. A calibration with gravimetrically prepared standard solutions was previously made to any determination of the equilibrium compositions. After adjust the parameters of the calibration equation a standard deviation in the mole fraction of ±0.002 was obtained.

3. Thermodynamic data

3.1. Binary vapor–liquid equilibrium data

Tables 2 and 3 list equilibrium temperature T , the liquid phase mole fraction of component i , x_i and the vapor phase mole fraction of component i , y_i for the binary systems 1-propanol (1) + ethylene glycol (3) and water (2) + ethylene glycol (3), respectively. In addition, activity coefficient, γ_i , has also been included. It was calculated assuming the non-ideality of both liquid and vapor phases from the following equation [4]:

$$\ln \gamma_i = \ln \frac{y_i P}{x_i P_i^0} + \frac{(B_{ii} - V_i^L)(P_i - P_i^0)}{RT} + \frac{P}{2RT} \sum_k y_k (2\delta_{ji} - \delta_{jk}) \quad (1)$$

where P is the equilibrium pressure, P_i^0 is the pure-component vapor pressure calculated by the equation and parameters reported in DIPPR tables [22] and listed in Table 4, V_i^L is the molar liquid volume of component i , calculated by the Rackett equation [23], B_{ii} and B_{jj} are the second virial coefficients of the pure gases, B_{ij} the cross second virial coefficient and

Table 1
Specifications of chemical samples.

Chemical name	Source	Purity (mass fraction)	Purification method	Analysis method
1-propanol	Aldrich	0.990	None	GC ^a
Water	Merck	0.995	None	GC ^a
Ethylene glycol	Merck	0.995	None	GC ^a

^a Gas chromatography.

Table 2Isobaric VLE data for 1-propanol (1)+ethylene glycol (3) binary system at 101.3 kPa.^a

T/K	x ₁	y ₁	γ ₁	γ ₃
470.27	0.000	0.000		1.000
441.64	0.033	0.632	2.623	0.994
426.73	0.062	0.786	2.379	0.979
414.35	0.106	0.874	2.089	0.964
407.91	0.140	0.909	1.954	0.948
402.50	0.175	0.931	1.879	0.945
398.17	0.217	0.946	1.749	0.951
395.72	0.245	0.953	1.674	0.951
392.32	0.301	0.962	1.528	0.969
389.32	0.362	0.970	1.397	0.972
385.92	0.445	0.977	1.282	0.994
384.07	0.487	0.981	1.253	1.002
381.92	0.553	0.985	1.189	1.015
379.63	0.648	0.989	1.096	1.067
378.40	0.681	0.990	1.093	1.101
377.57	0.709	0.991	1.082	1.113
376.35	0.756	0.992	1.059	1.194
375.42	0.795	0.993	1.042	1.385
373.73	0.860	0.995	1.024	1.584
371.07	0.970	0.998	0.957	4.028
370.40	1.000	1.000	1.000	

^a $u(T) = 0.02$ K, $u(p) = 0.1$ kPa, and $u(x_1) = u(y_1) = 0.002$.**Table 3**Isobaric VLE data for water (2)+ethylene glycol (3) binary system at 101.3 kPa.^a

T/K	x ₂	y ₂	γ ₂	γ ₃
470.27	0.000	0.000		1.000
439.27	0.101	0.687	0.924	1.000
426.56	0.164	0.813	0.932	1.000
419.34	0.212	0.862	0.929	0.997
415.85	0.238	0.890	0.932	0.995
411.91	0.276	0.915	0.937	0.991
407.62	0.318	0.927	0.939	0.989
403.10	0.367	0.952	0.941	0.985
399.9	0.407	0.964	0.946	0.977
395.92	0.464	0.968	0.953	0.970
391.87	0.526	0.976	0.955	0.967
388.43	0.592	0.987	0.960	0.965
385.20	0.656	0.991	0.970	0.957
382.19	0.724	0.994	0.972	0.948
379.99	0.781	0.996	0.990	0.941
377.57	0.85	0.997	0.993	0.918
375.46	0.921	0.999	0.998	0.851
374.32	0.959	0.999	0.998	0.823
373.15	1.000	1.000	1.000	

^a $u(T) = 0.02$ K, $u(p) = 0.1$ kPa, and $u(x_1) = u(y_1) = 0.002$.**Table 4**

Parameters of the vapor pressure equation.

Compound	A _i	B _i	C _i	D _i	E _i
1-propanol (1)	88.134	−8498.6	−9.0766	8.3303×10^{-18}	6
Water (2)	73.649	−7258.2	−7.3037	4.1653×10^{-6}	2
Ethylene glycol (3)	84.090	−10411.0	−8.1976	1.6536×10^{-18}	6

Parameters and vapor pressure equation obtained from DIPPR tables [21].

Vapor pressure equation: $\ln P^*(\text{Pa}) = A + B/T(\text{K}) + C \ln T(\text{K}) + D (T(\text{K}))^E$.

$$\delta_{ij} = 2B_{ij} - B_{jj} - B_{ii} \quad (2)$$

The standard state for the calculation of activity coefficients is the pure component at the pressure and temperature of the solution. Eq. (1) is valid from low to moderate pressures where the virial equation of state truncated after the second coefficient is adequate to describe the vapor phase of the pure components and their mixtures, and pure components in state liquid are considered incompressible over the pressure range under consideration. The

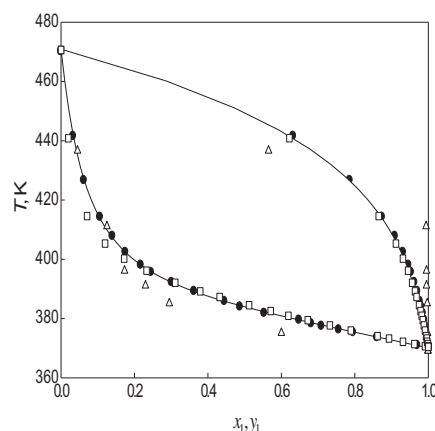


Fig. 1. Experimental and calculated VLE diagram for the 1-propanol (1)+ethylene glycol (3) binary system at 101.3 kPa. Experimental data: ●, from this work; Δ, data from Liu et al. [11]; □, data from Quian et al. [12]; calculated data: —, estimated by NRTL model with parameters from Table 6.

molar virial coefficients B_{ii} and B_{ij} were estimated by the method of Hayden and O'Connell [24] using the molecular parameters suggested by Prausnitz et al. [25].

Neither of the two binary systems form an azeotrope. It can be seen more clearly in Figs. 1 and 2. They show the variation of temperature with the composition of the two phases present in the binary system 1-propanol (1)+ethylene glycol (3) and water (2)+ethylene glycol (3), respectively. In order to compare the goodness of the acquired experimental values, data taken from reported works [13,14] have been included in Fig. 1. According to Fig. 1, only a few points of Liu et al. [13] match the experimental points. However, most of the points obtained by Qian et al. [14] coincide with the data of this work. The only significant deviations appear for low amounts of 1-propanol (<0.20 mole fraction). In Fig. 2 data taken from reported work have also been included. Once again data from Liu et al. [13] do not match with our experimental data. On the other hand, data from Kamihama et al. [15] fully agree with experimental data from this work.

The Fredenslund test [16] is a useful tool to check the thermodynamic consistency of the experimental data. It was

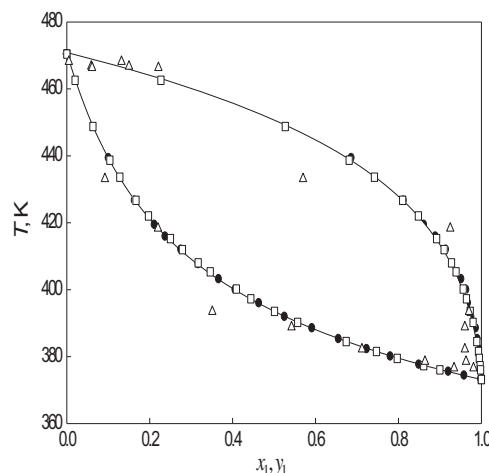


Fig. 2. Experimental and calculated VLE diagram for the water (2)+ethylene glycol (3) binary system at 101.3 kPa. Experimental data: ●, from this work; Δ, data from Liu et al. [11]; □, data from Kamihama et al. [13]; calculated data: —, estimated by NRTL model with parameters from Table 6.

Table 5

Consistency test statistics for the binary systems 1-propanol (1)+ethylene glycol (3) and water (2)+ethylene glycol (3).

System (i+j)	A ₁ ^a	A ₂ ^a	100 × AADy ₁ ^b	AADP ^c (kPa)
(1)+(3)	0.8406	−0.0866	0.349	0.456
(2)+(3)	−0.0271	−0.0056	0.162	0.338

^a Legendre polynomial parameters.

^b Average absolute deviation in vapor-phase composition.

^c Average absolute deviation in pressure.

applied to the binary data of this work and low deviations were obtained with a Legendre polynomial of two parameters. According to the results listed in Table 5, the absolute average deviation between experimental and calculated vapor phase composition have values lower than 0.005. Therefore, these data can be considered thermodynamically consistent. In addition to deviations, Table 5 also includes the two Legendre polynomial parameters of Fredenslund test.

The binary VLE data were correlated to obtain the parameters of the Wilson, NRTL and UNIQUAC local composition models. These parameters are listed in Table 6. The binaries interactions parameters of 1-propanol and water were taken from a previous work [7]. The rest of parameters were obtained after correlating VLE data by minimizing the following objective function (OF):

$$OF = \sum_{i=1}^N \left| \frac{T_i^{\text{exp.}} - T_i^{\text{calc.}}}{T_i^{\text{exp.}}} \right| + |y_i^{\text{exp.}} - y_i^{\text{calc.}}| \times 100 \quad (3)$$

where N is the total number of points and superscripts exp. and calc. are referred to experimental and calculated values, respectively. The molar liquid volumes of pure components required in Wilson model were calculated using the Rackett equation [23]. In the NRTL model, the randomness parameter α_{ij} was set to 0.3. This choice is in accordance with the recommendations of Prausnitz et al. [25]. The volume and surface parameters used in UNIQUAC model were taken from DECHEMA [26]. According to Table 6, the three models predict in a similar way the binary VLE data. Only in

Table 6

Parameters and deviations between experimental and calculated values for different excess Gibbs free energy models for the binary and ternary systems at 101.3 kPa.

Model	System i+j	A _{ij} J mol ^{−1}	A _{ji} J mol ^{−1}	α _{ij}	AADT ^a	AADy ₁ ^b	AADy ₂ ^b
Wilson ^c	(1)+(2) ^d	7057.10	4313.26				
	(1)+(3)	1785.41	1653.81	0.18	0.0058		
	(2)+(3)	2258.12	−862.21	0.55	0.0024		
	(1)+(2)+(3) ^e			1.49	0.0306	0.0307	
NRTL	(1)+(2) ^d	−435.76	8329.65	0.3			
	(1)+(3)	−99.3610	3435.653	0.3	0.22	0.0060	
	(2)+(3)	−1878.81	2147.35	0.3	0.15	0.0036	
	(1)+(2)+(3) ^e			0.47	0.0130	0.0157	
UNIQUAC ^f	(1)+(2) ^d	350.93	1498.39				
	(1)+(3)	87.7916	1325.798	0.31	0.0064		
	(2)+(3)	325.834	−1265.90	0.58	0.0036		
	(1)+(2)+(3) ^e			0.52	0.0169	0.0216	

^a Average absolute deviation in temperature.

^b Average absolute deviation in vapor phase composition.

^c Molar liquid volumes of pure components have been estimated with Rackett equation [22].

^d Parameters obtained previously in Pla-Franco et al. work [14].

^e Ternary estimation from binary parameters.

^f Volume and surface parameters from DECHEMA [25].

the water and ethylene glycol system the NRTL model shows a slight superiority. Anyway, the three models are able to accurately estimate the binary VLE data for the two binary systems.

3.2. Ternary vapor–liquid equilibrium data

In addition to the two binary systems, also VLE data for the ternary system 1-propanol (1)+water (2)+ethylene glycol (3) were obtained. The equilibrium temperature, the liquid and vapor phase composition and the activity coefficient of each component are listed in Table 7. Moreover, compositions of both phases are plotted on a ternary composition diagram in Fig. 3. The activity coefficients were calculated using Eq. (1) and the molar virial coefficients were estimated, as in the binary systems, by the method of Hayden and O'Connell [24]. The thermodynamic consistency was checked with the Wisniak and Tamir [17] modification of the McDermott–Ellis test [18]. This test is based on parameters D_a and $D_{\max,a}$, obtained as follows:

$$D_a = \sum_{i=1}^3 (x_{i_a} + x_{i_b}) (\ln \gamma_{i_a} - \ln \gamma_{i_b}) \quad (4)$$

Table 7

Isobaric VLE data for 1-propanol (1)+water (2)+ethylene glycol (3) ternary system at 101.3 kPa.^a

T/K	x ₁	x ₂	y ₁	y ₂	γ ₁	γ ₂	γ ₃
361.99	0.240	0.725	0.389	0.611	2.166	1.331	1.349
362.11	0.493	0.466	0.478	0.522	1.294	1.757	1.409
363.42	0.300	0.607	0.432	0.568	1.828	1.398	0.408
363.50	0.685	0.273	0.562	0.437	1.041	2.382	0.347
363.96	0.519	0.388	0.483	0.517	1.200	1.884	0.249
364.81	0.058	0.891	0.333	0.666	7.005	1.053	0.601
364.96	0.101	0.794	0.376	0.624	4.385	1.115	0.358
366.08	0.298	0.523	0.467	0.532	1.798	1.374	0.765
367.34	0.192	0.595	0.411	0.588	2.337	1.277	0.369
367.80	0.517	0.302	0.538	0.461	1.203	1.780	0.036
368.00	0.096	0.707	0.359	0.640	3.968	1.142	0.355
369.60	0.614	0.185	0.691	0.308	1.135	1.982	0.356
369.77	0.925	0.034	0.884	0.115	0.960	4.019	1.113
369.80	0.323	0.398	0.523	0.475	1.619	1.406	0.494
370.10	0.422	0.307	0.590	0.407	1.386	1.543	0.676
370.30	0.200	0.506	0.459	0.539	2.252	1.234	0.495
371.90	0.704	0.096	0.783	0.216	1.034	2.455	0.125
372.58	0.516	0.189	0.678	0.319	1.190	1.804	0.574
372.87	0.236	0.391	0.518	0.479	1.962	1.294	0.468
373.48	0.313	0.309	0.544	0.453	1.521	1.515	0.389
374.61	0.605	0.100	0.782	0.213	1.090	2.123	0.707

T/K	x ₁	x ₂	y ₁	y ₂	γ ₁	γ ₂	γ ₃
375.27	0.697	0.037	0.885	0.111	1.046	2.901	0.691
375.73	0.412	0.195	0.682	0.313	1.339	1.530	0.633
377.04	0.159	0.387	0.488	0.508	2.369	1.199	0.416
377.64	0.090	0.483	0.378	0.617	3.177	1.144	0.423
377.70	0.495	0.106	0.781	0.213	1.193	1.797	0.661
378.23	0.211	0.299	0.542	0.452	1.901	1.324	0.491
378.76	0.043	0.587	0.260	0.737	4.415	1.085	0.341
379.79	0.303	0.191	0.665	0.327	1.543	1.417	0.575
381.16	0.484	0.037	0.897	0.093	1.246	1.982	0.690
381.17	0.395	0.103	0.773	0.217	1.315	1.670	0.687
385.59	0.295	0.099	0.756	0.230	1.486	1.593	0.570
386.66	0.106	0.296	0.468	0.519	2.465	1.158	0.555
388.65	0.161	0.193	0.625	0.356	2.041	1.142	0.652
388.86	0.295	0.050	0.861	0.118	1.526	1.457	0.694
392.24	0.204	0.094	0.751	0.224	1.730	1.321	0.653
392.95	0.042	0.348	0.254	0.730	2.732	1.141	0.466
401.61	0.078	0.202	0.450	0.514	2.041	1.054	0.594
404.42	0.127	0.095	0.701	0.254	1.797	1.012	0.619

^a u(T) = 0.02 K, u(p) = 0.1 kPa, and u(x₁) = u(x₂) = u(y₁) = u(y₂) = 0.002.

$$D_{\max_a} = \sum_{i=1}^3 (x_{i_a} + x_{i_b}) \left(\frac{1}{x_{i_a}} + \frac{1}{y_{i_a}} + \frac{1}{x_{i_b}} + \frac{1}{y_{i_b}} \right) \Delta x$$

$$+ \sum_{i=1}^3 (x_{i_a} + x_{i_b}) \left(\frac{\Delta P}{P} \right) + 2 \sum_{i=1}^3 |\ln \gamma_{i_b} - \ln \gamma_{i_a}|$$

$$+ \sum_{i=1}^3 (x_{i_a} + x_{i_b}) B_j [(T_a + C_j)^{-2} + (T_b + C_j)^{-2}] \Delta T$$

in Eqs. (4) and (5) the subscripts *a* and *b* are referred to two consecutive experimental points. Δx , ΔP and ΔT are the standard uncertainty of the liquid phase mole fraction, the pressure and the temperature, respectively. To be thermodynamically consistent, D_a has to be smaller than $D_{\max_a}$ in all experimental points. As this condition is satisfied in this system, ternary VLE data can be considered thermodynamically consistent.

The model parameters obtained by correlating binary VLE data were used to estimate the ternary VLE data. Table 6 lists the average relative deviation between experimental and estimated temperatures. Wilson model presents certain difficulties to estimate correctly the temperature since the deviation obtained with this model is higher than the NRTL and UNIQUAC model. Table 6 also shows the average absolute deviations of the vapor phase mole fraction. Once again the Wilson model has the highest value of deviation value and the NRTL model provides the best predictions. For this reason, NRTL is chosen for all the subsequent theoretical calculations.

4. Process design

4.1. Distillation columns sequence

Residue curve maps (RCMs) are one of the most important tools to adequately establish a distillation columns sequence. They are the representation on a triangular composition diagram of residue curves. These curves describe the variation in a batch distillation of the system composition of the liquid phase. Fig. 4 shows the RCM for the 1-propanol (1)+water (2)+ethylene glycol (3) ternary system calculated with the NRTL model. Usually, the RCMs of azeotropic systems are divided into distillation regions. However, in this case there is only a distillation region. The binary azeotrope of 1-propanol and water is the unstable node and the ethylene

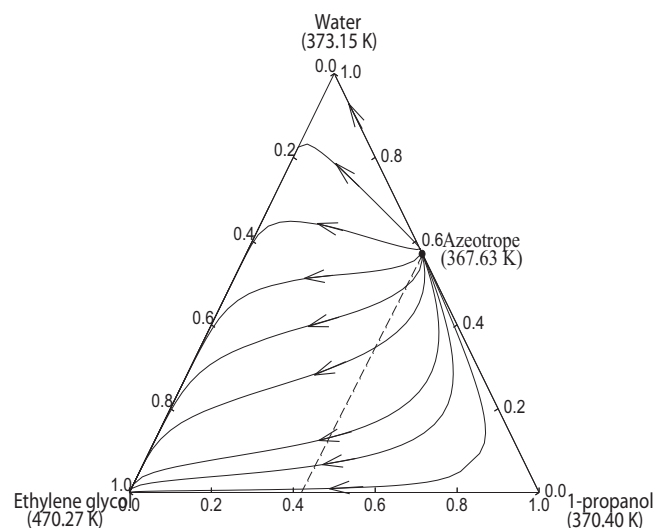


Fig. 4. Residue curve map for the ternary system 1-propanol (1)+water (2)+ethylene glycol (3) at 101.3 kPa built by Aspen Properties v 7.3 using the NRTL model with parameters given in Table 6: solid line, residue curve; long dash line, isovolatility line.

glycol is the stable node. The rest of vertices corresponding to pure components are saddle points.

The dashed line plotted in Fig. 4 is the isovolatility line. This line consists of all points that, in presence of the entrainer, have a relative volatility between 1-propanol and water equal to unity ($\alpha_{12}^S = 1$). According to Laroche et al. [27], if this line intersects the 1–3 edge, the more volatile component of the mixture, component (1), can be recovered as the top product of the first column in a two-columns sequence. This information is very useful to make an early design of the distillation sequence.

In this way, Fig. 5 shows the distillation columns sequence. The feed stream (F1), composed by 1-propanol and water, and the entrainer stream (E) enter the first distillation column (known as extractive column). In this column pure 1-propanol is obtained as overhead product (D1). The bottom stream (B1), mainly composed by water and ethylene glycol, is sent to the second distillation column (known as entrainer recovery column). This unit separates the water, obtained as top product (D2), and the ethylene glycol, obtained as the bottom product (B2). A certain amount of fresh ethylene glycol (make up) is added to B2 to form the entrainer stream, E, which is then recirculated to the extractive column.

In some cases the separation cannot be achieved with two columns and it takes an extra column. According to Laroche et al. [28], this situation occurs especially in cases where the curvature of the residue curves is not pronounced enough. A possible solution of this problem is that the feed stream enters a preconcentration column before the azeotropic column [29,30]. The main function of the preconcentration columns is distilling the mixture up to approximately the azeotropic point. In this work, the 1-propanol mole fraction of the feed stream, x_{1F} , was set at 0.40. This value is near to the azeotropic point and apparently no preconcentration column is needed.

4.2. Minimum amount of entrainer

The distillation columns sequence is not the only information provided by the isovolatility line; also specifies the minimum amount of entrainer required to break the azeotrope. In this way, the minimum mole fraction of entrainer, $x_{3 \min}$, is that placed in the isovolatility line when this line intersects with the edge of the more volatile component. In this case the isovolatility line

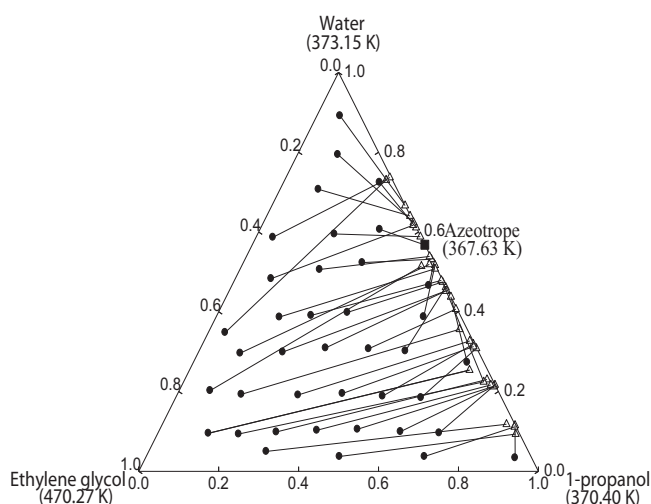


Fig. 3. Experimental VLE diagram for the 1-propanol (1)+water (2)+ethylene glycol (3) ternary system at 101.3 kPa: ●, liquid phase; △, vapour phase; ■, binary azeotrope.

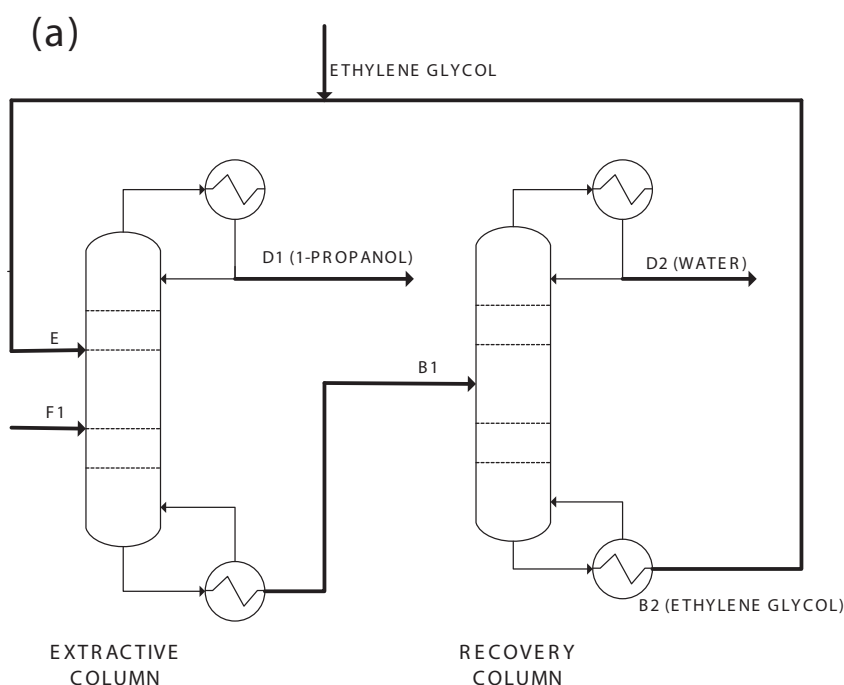


Fig. 5. General extractive distillation sequence for ethanol-water separation with a two-columns sequence.

intersects the 1-propanol (the more volatile component) edge at $x_{3\min} = 0.58$.

According to the isovolatility line, mixtures composed by at least 58% mol of ethylene glycol does not exhibit azeotropy. To check this, ternary VLE data were estimated for three fixed solvent compositions. The first solvent composition was adjusted to 0.4, a value below the minimum required to break the azeotrope. The second was 0.6, which is approximately the minimum required and the third solvent composition was 0.75, a value above the minimum. Data estimation was done with the NRTL model using the parameters listed in Table 6. Estimated compositions are plotted on a free-solvent diagram represented in Fig. 6. As expected, in the first case ($x_3 = 0.4$) the azeotrope does not disappear, although now the azeotropic composition is $x_1 = 0.69$ instead of 0.43. In contrast, the azeotrope has disappeared

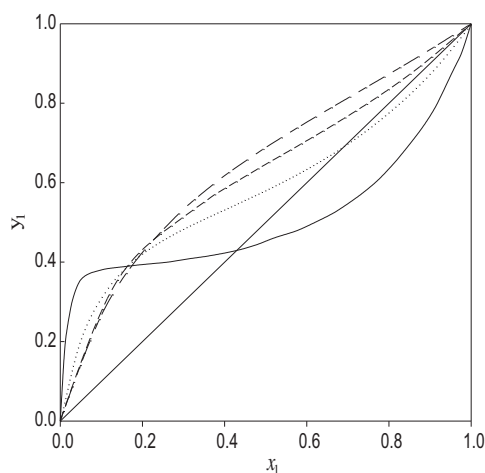


Fig. 6. VLE data plotted on a solvent-free basis for the system 1-propanol (1) + water (2) + ethylene glycol (3) at 101.3 kPa for $x_3 = 0.00$ (solid line), 0.40 (dotted line), 0.60 (short dashed line) and 0.75 (short-long dashed line). All values have been calculated using the NRTL model with the parameters given in Table 6.

with an entrainer mole fraction equal to 0.6. However, at high concentrations of 1-propanol the relative volatility α_{12}^S is too close to the unity to allow an adequate separation by distillation. This situation improves when the entrainer composition is equal to 0.75. Under this condition, the complete separation of 1-propanol and water does seem feasible.

5. Process simulation and optimization

5.1. Initial data

The sequence proposed above was simulated and optimized using Aspen Hysys v 8.0, a computer simulator developed by Aspen Technologies. The thermodynamic model used in the simulation was NRTL with the parameters listed in Table 6. As mentioned above, the 1-propanol mole fraction in the feed was set at 0.40. This value was chosen because it is close enough to the azeotropic point ($x_1 = 0.43$). The rest of process specifications were the mass flow and temperature of the initial streams, the pressure in each of the columns of the process and the purity and component recovery in both products. The values of these variables are listed in Table 8.

Table 8
Specification of the design variables in the extractive distillation process.

	Variables	Specification
Feed stream	Temperature (°C)	35
	Mass flow (kg/h)	1500
	Molar flow (kmol/h)	43.04
	1-propanol mass fraction	0.690
Extractive column	Operation pressure (kPa)	101.3
	Purity (1-propanol mass fraction)	0.995
	Recovery of 1-propanol (%)	99.0
Entrainment recovery column	Operation pressure (kPa)	101.3
	Purity (water mass fraction)	0.967
	Recovery of water (%)	99.0

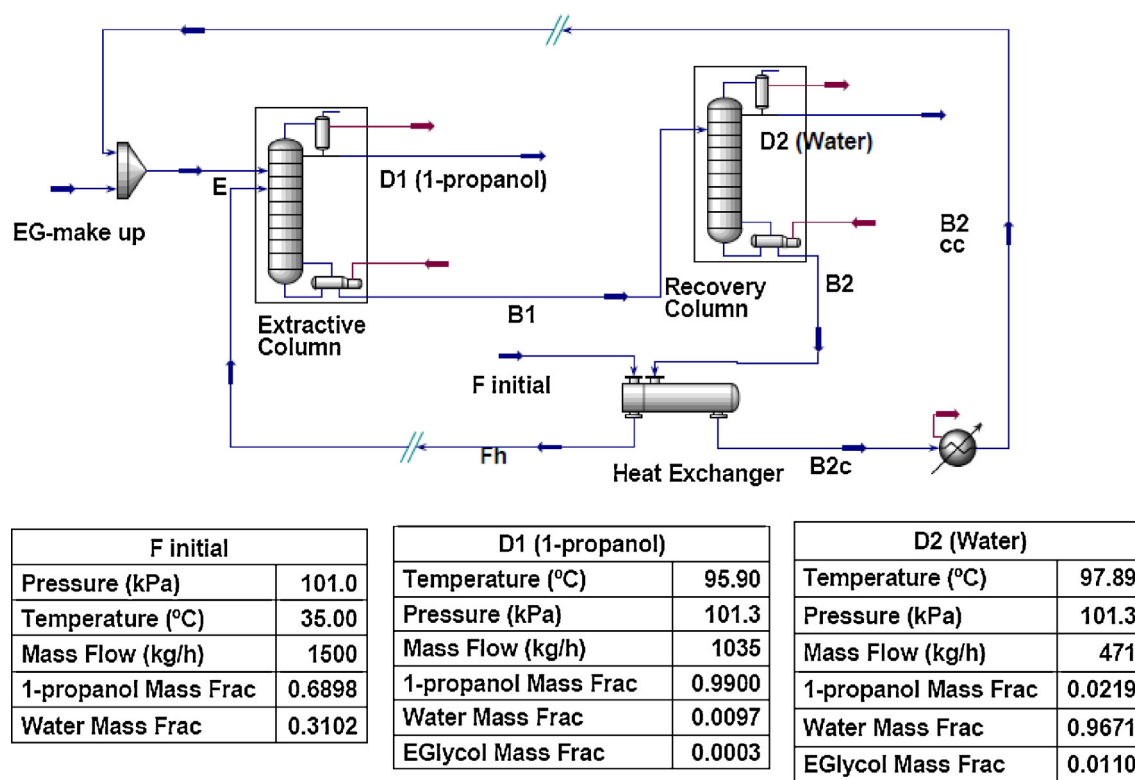


Fig. 7. Aspen Hysys process flow diagram (PFD) of extractive distillation sequence with a two-columns sequence.

5.2. Simulation and optimization

Fig. 7 illustrates the extractive distillation process. As can be seen, the initial mixture feed stream is heated in a heat exchanger before entering the extractive column to recover part of the energy expended in the reboiler of the entrainer recovery column. Thus, in this heat exchanger, the recovered entrainer stream is cooled and the feed stream is heated. Furthermore, this increasing in the temperature of the feed stream results in a decrease in the heat duty of the extractive column reboiler, as Soave and Feliu proved [31].

In order to obtain the optimal process variables, six cases were proposed and simulated. The initial difference between each case was the number of ideal trays in the extractive distillation column because, as mentioned above, it is the main unit of the process. Table 9 lists the number of ideal trays for each case. Once the total number of stages was specified, the next step was to obtain the entrainer process flow. In this way, the minimum value of entrainer flow capable of carrying out the separation according to requirements of Table 8 was calculated. However, the final value was set 15% above the minimum in all cases, since it is not recommended to work close to the limit values. Table 9 lists values of the molar

and mass entrainer-to-feed ratio. As expected, the case with the least number of stages, EC-1, has the highest entrainer-to-feed ratio. In addition, Table 9 also shows the optimal feed locations for each case, which is the location that provides the minimal reboiler heat duty (RHD). In this process, the entrainer should always enter the column at a stage near the top and the feed mixture stream should enter near the reboiler.

The other process column, the entrainer recovery column, was partially optimized using the reboiler heat duty (RHD) as a guideline. Since the column behavior is the same in all cases, calculations were made using data from EC-3 and then these results were extrapolated to the rest of cases. In this way, Fig. 8 shows the variation of the reboiler heat duty and the number of stages with respect to the reflux ratio. As in the previous column,

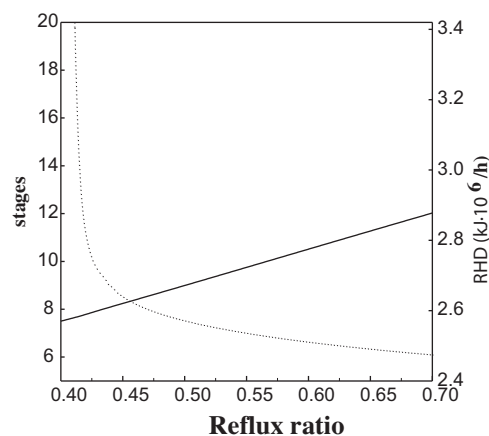


Fig. 8. Case study in the extractive distillation process. Variation of stage number (· · · · ·) and reboiler heat duty (—) as a function of the reflux ratio for the entrainer recovery column of EC-3.

Table 9
Specification of the design variables in the extractive column for each case.

	Case					
	EC-1	EC-2	EC-3	EC-4	EC-5	EC-6
Ideal trays number	16	20	25	30	40	50
Optimal feed stage ^a						
Initial mixture	3	3	3	3	3	3
Entrainment	12	15	20	25	32	40
Entrainment-to-feed ratio						
Molar	6.46	4.55	3.55	3.06	2.62	2.46
Mass	11.49	8.10	6.32	5.45	4.66	4.38

^a From top.

Table 10
Utility prices.^a

Utility	Price
Low-pressure steam (€/t)	10.00
Cooling water (€/m ³)	0.04
Electricity (€/kWh)	0.07

^a Consider Chemical Engineering Plant Cost Index (2013) = 567.3 and project life 10 years.

the location of the feed tray was optimized. According to Fig. 8, the intersection between the number of ideal trays and the RHD is around a reflux ratio of 0.475, which corresponds to a column of 8 theoretical stages. However, as it is desired to reduce energy consumption, the number of ideal stages was set to 10 and the feed stage was located at the 5th stage.

An economic evaluation was done for each case in order to find the best alternative. In this way, the total annual costs (TAC) [32] were calculated by:

$$TAC\left(\frac{\text{€}}{\text{year}}\right) = C_V + C_f + (i_m + i_r) \times FCI \quad (6)$$

where C_V is the process variable cost, which corresponds to utilities consumptions and was calculated using the utilities prices of Table 10; C_f is the annual fixed costs as maintenance and wages; FCI is the fixed investment; i_m is the minimum acceptable rate of return on FCI and i_r is the fixed capital recovery rate applied to FCI.

In this work, the annual fixed cost C_f was considered as the 10% of FCI; a typical value of 7% was taken to the minimum acceptable rate i_m and a value equal to 8.3% was assumed as the fixed capital recovery rate i_r . With these values, Eq. (6) can be rewritten as:

$$TAC\left(\frac{\text{€}}{\text{year}}\right) = C_V + 0.25 \times FCI \quad (7)$$

The software CAPCOST[®], developed by Turton et al. [33], was used to estimate the fixed capital investment. The Chemical Engineering Plant Cost Index (CEPCI) value was 567.3, corresponding to the year 2013. In the economic evaluation, the plant was considered to have a useful life of 10 years. Fig. 9 shows the estimated TAC for each case. According to the results, the minimum TAC is EC-4, which corresponds to an extractive column of 30 ideal

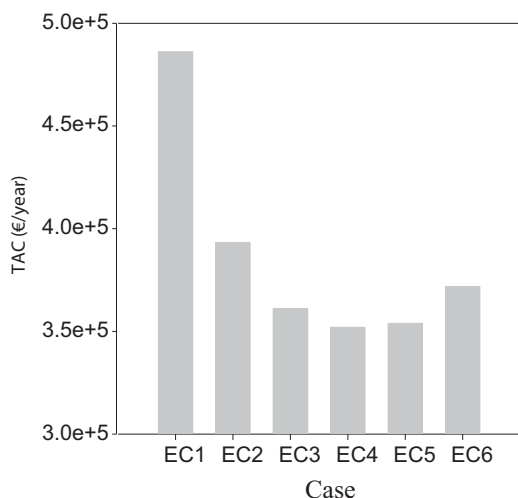


Fig. 9. Total annual cost (TAC) for the different cases in the extractive distillation.

Table 11

Results from simulation of the extractive distillation process for the 4th case (EC-4).

Variables	Value
Entrainer flow	
Make up flow (kg/h)	5.16
Overall flow (kg/h)	8179
Extractive column	
Ideal stages	30
Binary mixture feed stage	3
Entrainer feed stage	25
Assumed tray efficiency (%)	70
Reflux ratio	1.319
Heat duty reboiler (kJ/h)	1.778×10^6
Heat duty condenser (kJ/h)	1.707×10^6
Entrainer recovery column	
Ideal stages	10
Feed stage	5
Assumed tray efficiency (%)	70
Reflux ratio	0.426
Heat duty reboiler (kJ/h)	2.595×10^6
Heat duty condenser (kJ/h)	1.467×10^6
Heat exchanger	
Cold fluid ^a	
Inlet temperature (°C)	35.0
Outlet temperature (°C)	130.0
Hot fluid ^b	
Inlet temperature (°C)	199.2
Outlet temperature (°C)	105.0

^a The initial mixture stream.

^b The bottom product of the entrainer recovery column.

Table 12

Estimated capital investment of each individual unit in the extractive distillation process for the 4th Case (EC-4).

Unit	Cost (10 ³ €)
Extractive column	
Tower + trays	349.77
Reboiler	44.64
Condenser	3.75
Reflux pump	22.83
Reflux vessel	11.67
Total	432.65
Entrainer recovery column	
Tower + trays	116.67
Reboiler	94.73
Condenser	3.49
Reflux pump	13.02
Reflux vessel	6.00
Total	233.90
Heat exchanger	22.80
Cooler	19.74
Recycled pump	18.01
Fixed capital investment	727.19

stages and a TAC of 3.52×10^5 €/year. This value is slightly lower than values for EC-3 (3.61×10^5 €/year) and EC-5 (3.54×10^5 €/year). Table 11 lists values of the main process variables for the EC-4. As can be seen, the reflux ratio is 1.319 and 0.426 for the extractive and entrainer recovery column, respectively. For detailed information, Table 12 lists the estimated capital investment for each unit.

6. Conclusions

The use of ethylene glycol as entrainer for an extractive distillation to separate 1-propanol and water was proposed. In this way, the first part of the work was to acquire of isobaric VLE data at 101.3 kPa of the ternary system consisting of 1-propanol + water + ethylene glycol and its binary subsystems; 1-propanol + ethylene glycol and water + ethylene glycol. The experimental data were correlated with the Wilson, the NRTL and the UNIQUAC local

composition models and a set of binary interaction parameters was obtained for each model.

The second part was focused on the simulation and optimization of an extractive distillation process comprising two columns. Aspen Hysys 8.0[®] was used to carry out the simulation and the optimization. Thermodynamic calculations were done with the NRTL model parameters obtained in the previous step. Results showed that it is possible to use ethylene glycol to separate 1-propanol and water. The economical evaluation found an extractive column with 30 ideal stages and an entrainer-to-feed ratio of 3 as the best option for carrying out the process with ethylene glycol.

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